

PHYS 3717: Particle Physics

<https://homiller-hep.net/teaching/2026-2/>

Homework 2

Deadline: September 17, 2026 (in class)

Problem 1: Lorentz Tensors and Indices

Consider a tensor $A^{\mu\nu}$ and a vector B^μ , with components:

$$A^{\mu\nu} = \begin{pmatrix} 3 & 3 & 0 & -1 \\ -4 & 1 & 1 & 0 \\ 3 & -3 & 0 & 0 \\ 2 & 1 & 2 & -4 \end{pmatrix}, \quad B^\mu = \begin{pmatrix} -3 \\ 2 \\ 1 \\ 0 \end{pmatrix}. \quad (1)$$

Find the components of:

- (a) B_μ
- (b) $A^\mu{}_\nu$
- (c) $A_\mu{}^\nu$
- (d) $A^\lambda{}_\lambda$
- (e) $B^\mu B_\mu$
- (f) $B_\mu A^{\mu\nu}$

Problem 2: Lorentz Transformations

Here we'll work out some of the Lorentz transformation properties we mentioned but did not derive in class:

- (a) Starting from the condition that $\Lambda^T g \Lambda = g$, derive the following relations for a general Lorentz transformation $\Lambda^\mu{}_\nu$ (which could include boosts, rotations, and space or time reflections):

- (i) For the inverse transformations, $(\Lambda^{-1})^\mu{}_\nu \equiv \Lambda_\nu{}^\mu$, show that

$$\Lambda_\mu{}^\nu = g_{\mu\rho} \Lambda^\rho{}_\lambda g^{\lambda\nu} \quad (2)$$

and

$$\Lambda^\mu{}_\nu \Lambda_\mu{}^\rho = \Lambda_\nu{}^\mu \Lambda^\rho{}_\mu = \delta_\nu{}^\rho \quad (3)$$

- (ii) Show that the determinant satisfies

$$\det(\Lambda^\mu{}_\nu) = \pm 1 \quad (4)$$

(iii) Show that the element Λ^0_0 satisfies

$$(\Lambda^0_0)^2 \geq 1 \quad (5)$$

(b) Show that the four-gradient ∂_μ transforms as a covariant vector, i.e.,

$$\partial_\mu \equiv \frac{\partial}{\partial x^\mu} \mapsto \partial'_\mu \equiv \frac{\partial}{\partial x'^\mu} = \Lambda_\mu{}^\nu \frac{\partial}{\partial x^\nu} = \Lambda_\mu{}^\nu \partial_\nu. \quad (6)$$

Problem 3: Two-body Decay Kinematics

Consider the two-body decay $A \rightarrow B + C$, where the masses of the particles are m_A , m_B and m_C , respectively. Throughout this problem, we'll work in the rest frame of A , defined by $\vec{p}_A = 0$, $E_A = m_A c^2$.

(a) Using conservation of energy and conservation of momentum, show that the momenta (in magnitude), energies and velocities of B and C are given by

$$\begin{aligned} |\vec{p}_B| &= |\vec{p}_C| \equiv p = \frac{c}{2m_A} \sqrt{\lambda(m_A^2, m_B^2, m_C^2)}, \\ E_B &= \frac{c^2}{2m_A} (m_A^2 + m_B^2 - m_C^2), \quad E_C = \frac{c^2}{2m_A} (m_A^2 - m_B^2 + m_C^2), \\ v_B &= c \frac{\sqrt{\lambda(m_A^2, m_B^2, m_C^2)}}{(m_A^2 + m_B^2 - m_C^2)}, \quad v_C = c \frac{\sqrt{\lambda(m_A^2, m_B^2, m_C^2)}}{(m_A^2 - m_B^2 + m_C^2)}, \end{aligned} \quad (7)$$

where c is the speed of light, and we have defined the kinematic function,

$$\lambda(x, y, z) = x^2 + y^2 + z^2 - 2xy - 2xz - 2yz. \quad (8)$$

You are free to either keep the factors of c explicit throughout, or set $c = 1$ and then restore the correct factor at the end using dimensional analysis.

(b) The condition for the decay to be kinematically allowed is $p \geq 0$. Using your results from part (a), show that this condition implies the constraint

$$m_A \geq m_B + m_C \quad (9)$$

(c) Consider the special case $m_B \neq 0$ and $m_C = 0$ and obtain simplified expressions for the momenta (in magnitude), energies, and velocities of B and C directly from your results in part (a). As an application, consider the dominant decay of the charged pion,

$$\pi^- \rightarrow \mu^- + \bar{\nu}_\mu. \quad (10)$$

Visit <https://pdglive.lbl.gov> to find the masses of π^- and μ^- (consider the neutrino to be massless). Numerically compute the momenta (in units of MeV/ c), energies (in units MeV) and velocities (in units of c) of the muon and anti-neutrino.

- (d) Consider the special case $m_B = m_C \equiv m$, and obtain simplified expressions for the momenta, energies and velocities of B and C directly from your results in part (a). As an application, consider the dominant decay of the Higgs boson,

$$h \rightarrow b + \bar{b}, \quad (11)$$

where b is the bottom quark. Visit <https://pdglive.lbl.gov> to find their masses and numerically compute the momenta, energies and velocities in the appropriate units of the bottom / anti-bottom quarks.

Problem 4: Threshold Energies in Fixed Target Experiments

Here we'll work out the threshold energy for production of various particles in fixed target collisions.

- (a) Consider a process where particle A with mass m_A and energy E collides with particle B of mass m_B at rest, producing particles $C_1, C_2, \dots$ with masses $m_1, m_2, \dots$ through the reaction:

$$A + B \rightarrow C_1 + C_2 + \dots$$

Compute the minimum energy required for this reaction (the threshold energy) in terms of the masses of the particles.

- (b) Using your results from part (a), find the threshold energies for the following reactions, assuming the first particle listed in the initial state has energy E and the second particle listed in the initial state is at rest:

- (i) $\nu_\mu + n \rightarrow \mu^- + \pi^+ + n$,
- (ii) $p + p \rightarrow p + p + \pi^+ + \pi^-$,
- (iii) $\bar{\nu}_e + e^- \rightarrow W^-$.

Use <https://pdglive.lbl.gov> to find the masses of the various particles (you can take the neutrinos to be massless).